

IBM University Engagement Project Submission

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Domain of the Project : Education Technology

Project Title : Learn Mate AI – Personalized Course Pathway Agent using IBM Granite

Domain of the Project: Education Technology

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Problem Statement : Agentic AI for Personalized Course Pathways

The Challenge :

Students often struggle to identify the right learning path because of the overwhelming number of online courses and lack of personalized guidance.

- Many learners are confused about:
- what skills to learn
- which technologies are important
- which projects to build
- how to prepare for placements or career goals

The Objective :

Build an AI-powered Personalized Course Pathway Agent that:

- Analyzes student interests and skill levels
- recommends personalized learning roadmaps
- suggests relevant skills, courses, and projects
- provides career-focused guidance
- dynamically adapts recommendations

Technology Used:

Tech Stack I used:

Python + Streamlit + IBM Granite Concepts

Technologies Used:

- Python
- Stream lit
- JSON Knowledge Base
- IBM Granite Concepts
- Agentic AI Workflow

Proposed Solution :

The proposed solution is an AI-powered Personalized Course Pathway Agent that helps students identify the best learning roadmap based on their interests, skill levels, and career goals.

The system collects user inputs such as:

- ❖ preferred domain
- ❖ current skill level
- ❖ career objective

Based on this information, the AI system generates:

- ❖ recommended skills
- ❖ learning roadmap
- ❖ suggested courses
- ❖ mini projects
- ❖ career guidance

The solution uses Agentic AI concepts to create a dynamic and personalized educational assistant.

fusion and helping students focus on the right technologies and career path.

Langflow component Used :

1) Chat Input:

- ❖ Allows users to enter:
- ❖ interests
- ❖ skill level
- ❖ career goal

2) IBM Granite Concept:

Used for intelligent recommendation generation and personalized guidance

3) Recommendation Engine:

Analyzes user preferences and generates learning pathways

4) Chat Output:

- ❖ Displays:
- ❖ skills to learn
- ❖ courses
- ❖ projects
- ❖ career suggestions

5) File Component:

Stores roadmap and recommendation data using JSON knowledge base.

Lang flow workflow:

Workflow:

User Input



Interest & Skill Analysis



Recommendation Engine



Personalized Roadmap Generation



Course & Project Suggestions



Career Guidance Output



Personalized Course Pathway Agent

AI-powered personalized learning assistant for students.

Enter Your Name

T Rohitha

Choose Your Interest

Frontend Development

Select Skill Level

Beginner

Enter Your Career Goal

Data Analyst

Generate Roadmap

Role of Agentic AI in Learn Mate AI

Role of Agentic AI in Learn Mate AI

Agentic AI enables the system to function as an intelligent educational assistant instead of a basic chatbot.

The AI autonomously analyzes:

- user interests
 - skill levels
 - career goals
- and generates personalized learning pathways dynamically.

The system behaves like a smart mentor by:

- recommending technologies
 - suggesting projects
 - guiding career preparation
 - adapting recommendations
- This improves learning efficiency and helps students make better academic and career decisions.

Hello T Rohitha! Here is your personalized roadmap.



Skills to Learn ↔

- ✓ HTML
- ✓ CSS
- ✓ JavaScript



Recommended Courses

- Frontend Basics
- Responsive Web Design



Suggested Projects

- 🔥 Portfolio Website
- 🔥 Landing Page



Career Goal

Technology Used:

❖ **Streamlit:**

Used for creating the interactive web interface.

❖ **Python:**

Used for backend development and recommendation logic

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❖ **IBM Granite Concepts:**

Used for intelligent AI-based recommendation workflows.

❖ **JSON Knowledge Base**

Used to store roadmap and course recommendation data

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❖ **Agentic AI Workflow**

Used for dynamic personalized guidance generation.

Novelty and Uniqueness:

- ❖ Personalized Learning Roadmaps
- ❖ AI-powered Course Recommendations
- ❖ Career-focused Guidance
- ❖ Beginner-Friendly User Interface
- ❖ Dynamic Recommendation System
- ❖ Education-Focused Agentic AI Solution
- ❖ Project & Skill Suggestions in One Platform

Git Hub Link:

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Learn Mate AI Project Git hub Link:

<https://github.com/Rohitha970/LearnMate-AI.git>

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Future Scope

1. Integration with Online Learning Platforms:

The system can integrate with platforms like Coursera, Udemy, and IBM SkillsBuild to recommend real-time courses and certifications.

2. AI-Based Progress Tracking:

Future versions can monitor student progress and dynamically update learning pathways based on performance.

3. Resume & Placement Guidance:

The system can suggest resume improvements, mock interviews, and placement preparation strategies.

4. Multilingual Support:

Support for regional languages can improve accessibility for students from diverse backgrounds.

THANK YOU

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